AR-601M: two Jacobian derivations

Task 5, right and left arms

1. Model and velocity convention

Let $b = +1$ for the right arm and $b = -1$ for the left arm. Rotations and translations act on column vectors about or along the current local axis. The base-to-tool chain is

$$\begin{aligned} T(q) = {} & R_z(q_1)D_z(l_1)R_z(b\alpha)R_y(q_2)D_y(-bl_2) \\ & R_x(q_3)D_z(-l_3)R_z(q_4)R_y(q_5)R_x(q_6)D_x(l_4), \end{aligned} \tag{1}$$

where $\alpha = 10^\circ$ and $(l_1, l_2, l_3, l_4) = (0.40, 0.18, 0.30, 0.12)\,\mathrm{m}$. Write $T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix}$. We seek the *geometric* Jacobian of the tool origin: $v = \dot{p}$ is its linear velocity and $[\omega]_\times = \dot{R}R^\mathsf{T}$ is its angular velocity. Both are expressed in the base frame, and

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = J(q)\dot{q}, \qquad J \in \mathbb{R}^{6\times 6}. \tag{2}$$

Here $[x]_\times y = x \times y$, and $\mathrm{vee}([x]_\times) = x$. For joint i, let $P_i = \begin{bmatrix} U_i & p_i \\ 0 & 1 \end{bmatrix}$ be the prefix of the chain immediately *before* its rotation factor. Its local unit axis e_i follows the order z, y, x, z, y, x; its base-frame axis is $a_i = U_i e_i$. Fixed offsets and the $b\alpha$ rotation are included in P_i. Let $d_i \in \mathbb{R}^6$ denote the ith joint-space basis vector.

2. Geometric derivation: velocity about an axis

Hold the other joints fixed and increase only q_i by ε. The axis line passes through p_i in direction a_i, so the tool point and orientation change by

$$p(q + \varepsilon d_i) = p_i + \exp(\varepsilon[a_i]_\times)(p - p_i), \qquad R(q + \varepsilon d_i) = \exp(\varepsilon[a_i]_\times)R. \tag{3}$$

Using $\exp(\varepsilon[a_i]_\times) = I + \varepsilon[a_i]_\times + O(\varepsilon^2)$ and differentiating at $\varepsilon = 0$ gives

$$\frac{\partial p}{\partial q_i} = a_i \times (p - p_i), \qquad \frac{\partial R}{\partial q_i}R^\mathsf{T} = [a_i]_\times. \tag{4}$$

Because instantaneous contributions add, the ith column is

$$J_i^{\mathrm{geom}} = \begin{bmatrix} a_i \times (p - p_i) \\ a_i \end{bmatrix}, \qquad J^{\mathrm{geom}} = [J_1^{\mathrm{geom}} \ \cdots \ J_6^{\mathrm{geom}}]. \tag{5}$$

3. Independent derivation: differentiate the matrix product

Number every factor in the complete chain $T = F_1 \cdots F_N$; let factor F_{k_i} be joint i's pure rotation. With the local generator

$$G_i = \begin{bmatrix} [e_i]_\times & 0 \\ 0 & 0 \end{bmatrix}, \qquad F_{k_i}(q_i) = \exp(q_i G_i), \qquad \frac{\partial F_{k_i}}{\partial q_i} = G_i F_{k_i}, \tag{6}$$

the product rule, applied to the *full* chain, gives

$$\frac{\partial T}{\partial q_i} = \underbrace{F_1 \cdots F_{k_i - 1}}_{P_i} G_i F_{k_i} \underbrace{F_{k_i+1} \cdots F_N}_{S_i} = P_i G_i P_i^{-1} T. \tag{7}$$

Multiplication of the homogeneous blocks yields

$$P_i G_i P_i^{-1} = \begin{bmatrix} [a_i]_\times & -[a_i]_\times p_i \\ 0 & 0 \end{bmatrix}, \qquad \frac{\partial T}{\partial q_i} = \begin{bmatrix} [a_i]_\times R & [a_i]_\times (p - p_i) \\ 0 & 0 \end{bmatrix}. \tag{8}$$

Thus the derivative of the translation gives the top three entries, while the derivative of the rotation gives the bottom three:

$$J_i^{\mathrm{prod}} = \begin{bmatrix} \partial p/\partial q_i \\ \mathrm{vee}((\partial R/\partial q_i)R^\mathsf{T}) \end{bmatrix} = \begin{bmatrix} a_i \times (p - p_i) \\ a_i \end{bmatrix}. \tag{9}$$

The matrix inside vee is skew-symmetric: differentiating $RR^\mathsf{T} = I$ gives $(\partial R/\partial q_i)R^\mathsf{T} + R(\partial R/\partial q_i)^\mathsf{T} = 0$. Equations (5) and (9) therefore derive the same base-frame Jacobian by independent routes. Both are implemented in `src/task5_jacobian.py`; centered FK differences and pairwise residuals are reported in `REPORT.md`, Task 5.